

Pixels to Predictions: QLoRA Fine-Tuning of SmolVLM for Multimodal Science Multiple-Choice Reasoning

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Abstract

This report describes my final system for the *Pixels to Predictions: DL Vision Challenge*, a multimodal scientific multiple-choice reasoning task. Each example contains an image, a question, answer choices, optional hint and lecture text, and pedagogical metadata. The goal is to predict the correct 0-indexed answer choice for each hidden test example. My final system uses the official starter-notebook base model, HuggingFaceTB/SmolVLM-500M-Instruct, with QLoRA adaptation. The model is loaded in 4-bit NF4 precision, and only LoRA adapter parameters are trained. The final LoRA setup uses rank 8, alpha 16, dropout 0.05, and approximately 4.78M trainable parameters. For inference, I use deterministic multiple-choice likelihood scoring instead of free-form generation. The best final checkpoint, obtained after cautious continued fine-tuning to epoch 4, achieved a public leaderboard score of **0.86317**. The main improvements came from task-specific QLoRA fine-tuning, stable checkpoint continuation, manual context shortening, avoiding image-token truncation, and likelihood-based answer selection.

1 Introduction

The Pixels to Predictions challenge evaluates multimodal scientific reasoning. Each input example contains an image, textual context, a question, and several answer choices. The model must combine the visual evidence with the text to select the correct answer. This is harder than ordinary text-only classification because the answer may depend on diagrams, natural images, maps, charts, or visual clues that are not fully captured in the question text.

The objective is to output one valid 0-indexed answer choice for each hidden test example. The leaderboard metric is classification accuracy. Since the task is multiple-choice, the model should not produce open-ended explanations or arbitrary generated text. It should select one of the valid choices.

My final approach uses the official SmolVLM base model from the starter notebook, HuggingFaceTB/SmolVLM-500M-Instruct. I adapt the model using QLoRA because full fine-tuning is too expensive under free-tier Colab/Kaggle constraints. For inference, I score each valid answer option directly using model likelihood and choose the highest-likelihood option. The final system reached a public leaderboard score of **0.86317**.

The main contributions of this solution are:

- a reproducible QLoRA setup under the trainable-parameter constraint;
- a compact multimodal prompt format using image, metadata, lesson, hint, question, and choices;
- manual context shortening to avoid SmolVLM image-token truncation errors;
- deterministic likelihood-based multiple-choice inference;

- checkpoint-based recovery and continuation for unstable Colab sessions;
- ablations showing the effect of frozen inference, QLoRA fine-tuning, continued training, learning-rate continuation, and prompt/inference variants.

2 Dataset

2.1 Dataset Description

The dataset is a multimodal science question-answering dataset. It contains train, validation, and test splits. Each row corresponds to one multiple-choice question instance.

The important fields are:

- id: unique example identifier,
- image_path: relative path to the associated image,
- question: natural-language question,
- choices: JSON list of candidate answers,
- num_choices: number of valid answer choices,
- answer: 0-indexed correct answer for train and validation,
- hint: optional short supporting context,
- lecture: optional longer background passage,
- task, grade, subject, topic, category, and skill: metadata.

The final cleaned notebook loaded 3,109 training examples, 1,048 validation examples, and 1,008 test examples. For final leaderboard training, I used the combined train and validation set of 4,157 labeled examples. The validation split was still used during development to select the final method, prompt format, and inference approach.

Split	Examples	Labels Available
Train	3,109	Yes
Validation	1,048	Yes
Train + Validation	4,157	Yes
Test	1,008	No

Table 1: Dataset split sizes used in the final notebook.

2.2 Dataset Statistics

Figure 1 shows the dataset plots generated from my final clean notebook. The training answer distribution is not perfectly uniform. Answer indices 0 and 1 appear most often, answer 2 is also common, and answer 4 is rare. The subject distribution is dominated by natural science, followed by social science and language science. The number-of-choices distribution shows that 3-choice questions are most common, followed by 4-choice and 2-choice questions.

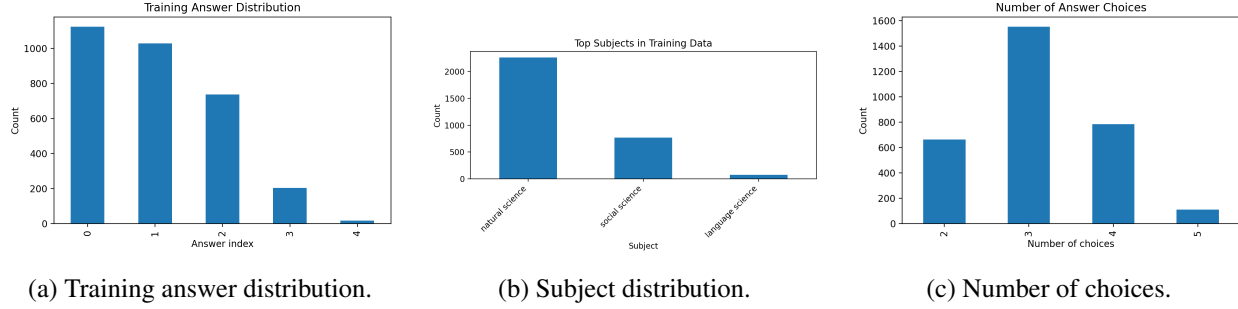


Figure 1: Dataset statistics generated in the final clean notebook. These figures motivated using valid-choice likelihood scoring instead of assuming a fixed number of answer options.

The exact train/test choice-count distribution is shown in Table 2. Because the number of choices varies, the inference code must score only valid options for each row. I therefore use the `num_choices` field during scoring and validate that each final answer is within the allowed range.

Number of Choices	Train Count	Test Count
2	664	272
3	1552	438
4	783	260
5	110	38

Table 2: Distribution of answer choice counts in train and test.

2.3 Preprocessing

The preprocessing pipeline is intentionally simple and reproducible:

- parse the choices field from JSON strings into Python lists;
- load images as RGB;
- resize images to 384×384 ;
- safely handle missing or optional text fields;
- manually shorten long lecture and hint fields;
- preserve `num_choices` so only valid options are scored;
- validate that final predictions contain no missing IDs and no invalid answer indices.

No external dataset was used. All training, validation, and inference were done only with the provided Kaggle competition data.

3 Model Description

3.1 Base Model

The base model is HuggingFaceTB/Smo1VLM-500M-Instruct. This model was selected because it is the official starter-notebook model, supports image-text reasoning, and is small enough to use under free-tier

GPU constraints. It is also within the model-size guidance for the competition.

3.2 QLoRA Adaptation

The final system uses QLoRA-style adaptation. The base model is loaded in 4-bit NF4 precision using bitsandbytes. Only LoRA adapter parameters are trained, while the base model weights remain frozen. This allows the model to adapt to the competition format without full fine-tuning.

The final LoRA configuration is:

- LoRA rank: $r = 8$,
- LoRA alpha: 16,
- LoRA dropout: 0.05,
- bias: none,
- task type: causal language modeling,
- target modules: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj.

This produced approximately 4.78M trainable parameters during fine-tuning, which stays below the 5M trainable-parameter limit used during development.

Component	Final Setting
Base model	SmolVLM-500M-Instruct
Adaptation	QLoRA / LoRA adapters
Quantization	4-bit NF4
LoRA rank	8
LoRA alpha	16
LoRA dropout	0.05
Target modules	q/k/v/o + gate/up/down projections
Trainable parameters	~4.78M
Image size	384×384
Final checkpoint	Epoch 4 cautious continuation
Inference	Multiple-choice likelihood scoring
Public score	0.86317

Table 3: Final model and inference configuration.

3.3 Justification of Design Choices

The final design was chosen for performance, reproducibility, and compute feasibility. SmolVLM is small enough to run on a Tesla T4 but still has useful multimodal reasoning ability. QLoRA allows task adaptation while staying under memory and trainable-parameter limits. Applying LoRA to the language-model attention and MLP projections adapts the reasoning and answer-selection components without modifying the full vision-language model.

Likelihood scoring was chosen because the task requires selecting one of a fixed set of answer choices. It prevents invalid generated outputs and makes inference deterministic.

4 Prompt and Data Pipeline

4.1 Prompt Format

Each example is converted into a compact multimodal prompt. The prompt starts with the <image> token, followed by metadata, shortened context, question, and answer choices. The final prompt format is:

```
<image>
Answer this science multiple-choice question. Reply with only the option letter.
Grade: ...
Subject: ...
Topic: ...
Category: ...
Skill: ...
Lesson: ...
Hint: ...
Question: ...
Choices:
A. ...
B. ...
C. ...
Answer:
```

I used option letters rather than raw numeric answers for likelihood scoring because the answer portion is short, explicit, and easy to mask.

4.2 Manual Context Shortening

A major implementation detail was avoiding processor-level truncation with image inputs. During development, using `truncation=True` with `SmolVLM` caused image-token mismatch errors. The final pipeline manually shortens text fields before tokenization:

- lecture: first 1200 characters,
- hint: first 700 characters.

The processor is then called with `truncation=False`. This keeps the image tokens intact while controlling memory usage.

5 Inference Method

5.1 Multiple-Choice Likelihood Scoring

The final system does not use free-form generation. Instead, it scores every valid answer option.

For a question with n choices, I construct n candidate inputs:

$$x_i = \text{prompt} + \text{option}_i,$$

where $i \in \{1, \dots, n\}$.

The model computes token-level cross-entropy loss for each candidate. The prompt portion is masked out, and the answer portion is scored. The score for option i is:

$$s_i = -\frac{1}{|A_i|} \sum_{t \in A_i} \ell_t,$$

where A_i is the set of answer tokens and ℓ_t is the token-level loss. The predicted answer is:

$$\hat{y} = \arg \max_i s_i.$$

This method is deterministic under `model.eval()` and `torch.no_grad()`.

5.2 Why Likelihood Scoring Instead of Generation

Free-form generation can produce extra words, invalid formatting, or an answer outside the valid choice set. Likelihood scoring avoids these problems because it only compares valid options. It also handles 2-, 3-, 4-, and 5-choice questions naturally.

6 Experimentation

6.1 Training Setup

The final training setup used:

- model: HuggingFaceTB/SmolVLM-500M-Instruct,
- optimizer: AdamW,
- batch size: 1,
- gradient accumulation: 8,
- image size: 384×384 ,
- final data: train + validation,
- quantization: 4-bit NF4,
- LoRA rank: 8,
- LoRA alpha: 16,
- LoRA dropout: 0.05.

Adapters and submissions were saved after important checkpoints because Colab runtime disconnections were a real issue. This made the final result recoverable and reproducible.

6.2 Development Issues and Fixes

Several engineering fixes were important:

- **Package mismatch:** `transformers==4.40.2` did not correctly instantiate the SmolVLM processor. The final environment uses `transformers==4.57.1`.
- **4-bit loading:** removing `bitsandbytes` broke the memory-efficient 4-bit setup. The final environment keeps `bitsandbytes`.
- **Image-token mismatch:** processor truncation caused image-token count errors. The final pipeline manually shortens text and sets processor truncation to false.

Experiment	Method	Public Score	Main Change	Observation
Frozen SmolVLM baseline	No fine-tuning; likelihood scoring only	0.58551	Base model only	Useful zero-shot ability, but weak adaptation to the contest answer format.
QLoRA epoch 2	4-bit SmolVLM + LoRA $r = 8$ on train+validation	0.80080	Supervised adapter training	Large improvement from task-specific fine-tuning.
Continued epoch 3	Reloaded adapter, fixed gradient issue, continued training	0.85510	Stable checkpoint continuation	Major improvement after fixing training stability.
Cautious epoch 4	Lower-learning-rate continued training	0.86317	Smaller LR continuation	Best final result.
Epoch 5 partial	Extra low-LR partial fine-tuning	0.86317	Additional small update	No measurable improvement over epoch 4.
Prompt ensemble / text fallback	Prompt variants and TF-IDF/text fallback	–	Inference-time blending	Weaker prompt variants reduced validation behavior, so this was not used in final.

Table 4: Main experiments and ablations.

- **PEFT reload gradients:** after loading a saved adapter, some resumed training attempts produced a loss without gradients. Disabling gradient checkpointing and explicitly enabling LoRA parameters fixed this.
- **Runtime instability:** saving adapters and submissions after important epochs allowed recovery after disconnects.

7 Results

7.1 Main Results

The final public leaderboard score is **0.86317**. Table 4 summarizes the main experiments.

7.2 Leaderboard Score Trend

Figure 2 shows the main public leaderboard score progression. The frozen SmolVLM likelihood baseline reached 0.58551. QLoRA fine-tuning improved performance to 0.80080 by epoch 2. Continued training with the fixed PEFT reload setup reached 0.85510 at epoch 3. The best result came from cautious epoch-4 continued training with a public score of 0.86317. A partial epoch-5 continuation did not improve further.

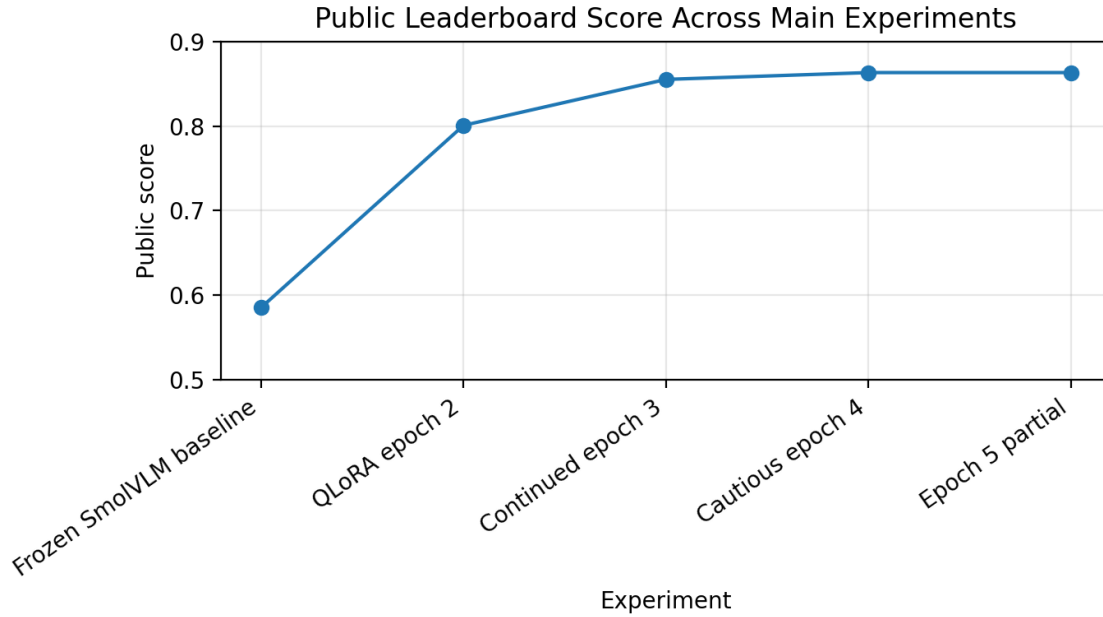


Figure 2: Public leaderboard score across the main experiments.

7.3 Analysis of Improvements

The largest gain came from QLoRA fine-tuning. The frozen baseline scored 0.58551, while the QLoRA epoch-2 model reached 0.80080. This shows that the base model has useful multimodal knowledge but needs task-specific adaptation.

The next major gain came from stable continued training. After fixing the PEFT reload gradient issue, epoch 3 reached 0.85510. Cautious epoch-4 continuation improved to 0.86317. The epoch-5 partial run tied epoch 4, suggesting the model had mostly saturated under this setup.

7.4 Final Submission Distribution

The final submission had 1,008 rows and two columns: id and answer. The regenerated final prediction distribution was:

- answer 0: 359 examples,
- answer 1: 368 examples,
- answer 2: 207 examples,
- answer 3: 71 examples,
- answer 4: 3 examples.

This distribution is consistent with the fact that only 38 test examples have 5 choices, so option 4 is valid for relatively few rows.

8 Ablations and Experimental Rigor

8.1 Number and Relevance of Ablations

I tested multiple relevant ablations:

1. frozen SmolVLM likelihood scoring;
2. QLoRA fine-tuning with rank 8;
3. resumed training after epoch 2;
4. lower-learning-rate continuation;
5. partial epoch-5 continuation;
6. prompt variants;
7. text fallback / TF-IDF-style fallback;
8. manual context shortening instead of processor truncation.

These ablations are relevant because they test the main design decisions: whether to fine-tune, how to continue training, how to infer answers, and how to handle long context.

8.2 Control and Fairness of Comparisons

The main comparable experiments use the same base model, same dataset, same prompt family, and same likelihood-scoring inference method. This makes the frozen baseline, QLoRA epoch 2, continued epoch 3, cautious epoch 4, and epoch 5 partial results fair to compare.

Some exploratory experiments changed the inference procedure, such as prompt ensemble or text fallback. I did not treat those as final improvements because they weakened validation behavior and were not directly comparable to the main checkpoint progression.

8.3 Analysis Quality

The score trend gives a clear interpretation. Fine-tuning is necessary, stable continuation helps, and extra training eventually saturates. The system improved from 0.58551 to 0.86317 through controlled changes. The final chosen model is not only the highest-scoring checkpoint, but also the most reproducible one because it is saved as a standalone LoRA adapter and can regenerate the final submission.

9 Code and Reproducibility

9.1 Repository Organization

The final repository should be organized as:

```
pixels-to-predictions-sv2881/  
  README.md  
  requirements.txt  
  notebooks/  
    sv2881_pixels_to_predictions_final_clean.ipynb
```

```
submissions/  
    submission_final_epoch4_086317.csv  
results/  
    ablation_results.csv  
    public_score_trend.png  
figures/  
    answer_distribution_train.png  
    subject_distribution_train.png  
    num_choices_distribution.png  
report/  
    final_report.pdf
```

The clean notebook is organized into environment setup, data loading, dataset visualization, prompt formatting, model loading, likelihood scoring, submission generation, ablations, and reproducibility notes. This avoids submitting a messy development notebook with failed cells and repeated recovery attempts.

9.2 Reproducible Steps for README

The README should include these exact rerun steps:

1. Clone the repository.
2. Install dependencies from `requirements.txt`.
3. Download or mount the Kaggle competition data in the expected folder.
4. Download the final LoRA adapter from the provided model-weights link.
5. Open `notebooks/sv2881_pixels_to_predictions_final_clean.ipynb`.
6. Run the environment setup cell.
7. Run the inference cells to regenerate `submission_final_epoch4_086317.csv`.
8. Upload the generated CSV to Kaggle.

9.3 Environment

The reproducible environment is:

```
transformers==4.57.1  
tokenizers==0.22.1  
huggingface_hub==0.36.0  
accelerate==1.11.0  
peft==0.13.2  
bitsandbytes  
datasets  
pillow  
tqdm  
matplotlib  
pandas  
numpy  
torch
```

9.4 Reproducible Training Setup

The report and notebook specify:

- fixed package versions,
- random seed 777,
- official SmolVLM checkpoint,
- LoRA rank 8 / alpha 16 / dropout 0.05,
- target modules,
- image size 384,
- train+validation use for final adapter,
- saved checkpoint path,
- final adapter link.

9.5 Reproducible Inference Setup

The final inference procedure:

- loads the saved epoch-4 LoRA adapter,
- calls `model.eval()`,
- uses `torch.no_grad()`,
- scores each valid answer option by likelihood,
- chooses the highest-scoring option,
- validates test IDs and answer ranges,
- saves a CSV with exactly id and answer.

9.6 Links

GitHub repository: [PASTE_GITHUB_REPO_LINK_HERE](#)

Model weights / LoRA adapter: [PASTE_MODEL_WEIGHTS_LINK_HERE](#)

Final submission file: `submission_final_epoch4_086317.csv`

Replace the two placeholder URLs before final submission.

10 Error Analysis and Limitations

10.1 Engineering Error Analysis

The main engineering failure modes were:

- **Package mismatch:** older transformers versions could not load the SmolVLM processor;
- **4-bit setup issue:** removing bitsandbytes broke memory-efficient loading;
- **Image-token mismatch:** processor-level truncation caused image-token count errors;
- **Runtime disconnects:** Colab sessions disconnected during long runs;
- **No-gradient reload issue:** PEFT adapter reloads sometimes produced a loss without a gradient path.

The final solution fixes these problems by pinning package versions, keeping bitsandbytes, manually shortening context, saving checkpoints, disabling gradient checkpointing during resumed training, and using a clean final inference notebook.

10.2 Model Limitations

The model may still struggle with:

- fine-grained diagrams with small visual details;
- chart or graph reading;
- exact spatial reasoning;
- long scientific passages where the key evidence appears late;
- rare 5-choice examples;
- examples where the image and text contain subtle conflicting clues.

A multi-seed LoRA ensemble or checkpoint averaging approach might improve private leaderboard robustness, but those experiments were limited by runtime constraints.

11 AI Tooling Disclosure

I used ChatGPT as a coding and debugging assistant. It helped with Colab setup, package compatibility, QLoRA debugging, checkpoint recovery, image-token truncation debugging, and report organization. All training, inference, and leaderboard submissions used only the provided Kaggle competition data. No external training data was used.

12 Conclusion

This project shows that a compact vision-language model can perform strongly on multimodal science multiple-choice reasoning when adapted carefully. The final system uses SmolVLM-500M-Instruct with QLoRA and deterministic multiple-choice likelihood scoring. The best checkpoint achieved a public leaderboard score of **0.86317**.

The most important findings were:

- QLoRA fine-tuning is much stronger than frozen inference;
- likelihood scoring is more reliable than free-form generation;
- manual context shortening is safer than processor truncation for SmolVLM;
- stable checkpoint continuation improves performance;
- extra training eventually saturates.

Future work could explore multi-seed ensembles, validation-based checkpoint averaging, better prompt selection, and category-level error analysis.

References

- [1] Hugging Face. HuggingFaceTB/SmolVLM-500M-Instruct. Official pretrained checkpoint used in the starter notebook.
- [2] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models.
- [3] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient Finetuning of Quantized LLMs.
- [4] Tim Dettmers et al. bitsandbytes: 8-bit and 4-bit CUDA quantization utilities.

A Appendix A: Rubric Alignment

A.1 Methodology

Sections 1–5 define the task, describe the dataset, explain the model, detail the data pipeline, and justify QLoRA plus likelihood scoring.

A.2 Code and Reproducibility

Section 10 specifies the repository structure, package versions, README rerun steps, final adapter, final submission, and deterministic inference procedure.

A.3 Ablations and Experimental Rigor

Sections 8 and 9 include frozen baseline, QLoRA epoch 2, continued epoch 3, cautious epoch 4, epoch 5 partial, prompt ensemble, text fallback, and truncation/preprocessing ablations. The comparisons are separated into controlled checkpoint comparisons and exploratory inference variations.

A.4 Error Analysis

Section 11 discusses engineering errors and model limitations, including package mismatch, image-token truncation, runtime instability, PEFT reload gradients, diagram reading, spatial reasoning, long context, and rare 5-choice cases.